



Frequently Asked Data Engineer Interview Questions (0–5 Years)

SQL (Most Common Round 1)

1. What is the difference between INNER JOIN, LEFT JOIN, RIGHT JOIN, and FULL OUTER JOIN?
2. How do you remove duplicates from a table without using DISTINCT?
3. Write a query to find the 2nd highest salary.
4. Explain window functions with an example.
5. How do you handle NULL values in SQL?
6. Difference between WHERE vs HAVING.
7. Write a query to find customers who placed more than 3 orders in the last 7 days.
8. How to pivot and unpivot data?
9. Explain primary key, unique key, and composite key.
10. How to check the performance of a SQL query?

Data Modeling / ETL Basics

11. Difference between OLTP vs OLAP.
12. What are fact and dimension tables?
13. Explain star schema vs snowflake schema.
14. What is SCD? Types of SCD?
15. Difference between batch vs real-time processing.
16. What is a pipeline? Explain the end-to-end workflow.

- 17. What is data partitioning and when do you use it?
- 18. What is data lineage?
- 19. What is data profiling?
- 20. What is normalization vs denormalization?

Python for Data Engineering

- 21. How do you handle large files in Python?
- 22. Difference between list, tuple, dictionary, and set.
- 23. How do you handle exceptions in Python?
- 24. How to read/write JSON, CSV, Parquet using Python?
- 25. What is multiprocessing vs multithreading?

Spark (Very Common 0–5 YOE)

- 26. Difference between RDD, DataFrame, Dataset.
- 27. What is lazy evaluation in Spark?
- 28. What is Spark partitioning?
- 29. How do you remove duplicates in PySpark?
- 30. Explain wide vs narrow transformations.
- 31. What is a shuffle in Spark?
- 32. What is broadcast join?
- 33. What is Spark caching/persistence?
- 34. How does Spark handle fault tolerance?
- 35. How do you optimize a Spark job?

Data Warehousing

- 36. Explain ETL vs ELT.
- 37. What is a data mart?
- 38. What is CDC (Change Data Capture)?
- 39. What is a surrogate key?
- 40. What is the difference between incremental load vs full load?

Cloud (AWS / Azure / GCP)

- 41. Difference between AWS S3 vs Azure Blob.
- 42. What is a data lake vs data warehouse?
- 43. What is serverless computing?
- 44. What is IAM and why is it used?
- 45. What are VPC, subnet, and security groups?
- 46. Explain Glue / Data Factory / Dataflow (based on your cloud).
- 47. What is EMR / Databricks / HDInsight?

- 48. What is an orchestrator?
- 49. What is autoscaling?
- 50. What is monitoring and logging? (CloudWatch, Log Analytics, Stackdriver)

Kafka / Streaming

- 51. What is Kafka?
- 52. What is a topic, partition, and offset?
- 53. Difference between consumer group vs consumer.
- 54. What is at-least-once vs at-most-once vs exactly-once delivery?
- 55. What is schema registry?

System Design for Data Engineering

- 56. How do you design a data pipeline?
- 57. How do you handle schema evolution?
- 58. How to design a deduplication system?
- 59. What is orchestration vs automation?
- 60. How do you ensure data quality?